

Week 1:

A local library is currently managing its books and members using a physical notebook, which has resulted in several operational problems such as misplaced books, difficulty tracking borrowed items, and inaccurate member records. To modernize its operations, the library plans to develop a database system called BookFlow. As a database developer, your task is to create the SQL foundation for this system. First, create a new database named `bookflow_db3`. Then, design two tables: a `Books` table containing the columns `book_id`, `title`, `isbn`, and `published_year`, and a `Members` table containing the columns `member_id`, `full_name`, and `email`. Ensure data integrity by setting `book_id` and `member_id` as PRIMARY KEYS, applying a NOT NULL constraint to the `title` column so that every book must have a name, enforcing UNIQUE constraints on both `isbn` and `email` to prevent duplicate entries, and adding a CHECK constraint to ensure that the `published_year` is not in the future (for example, it must be less than 2027). Finally, insert three sample records into both the `Books` and `Members` tables to verify that the database structure and constraints are functioning correctly. This exercise will help you develop skills in schema definition, data integrity enforcement, and the use of SQL Data Definition Language (DDL) commands such as `CREATE DATABASE`, `CREATE TABLE`, and `INSERT` statements.

Solutions:

```
mysql> create database bookflow_db3;
Query OK, 1 row affected (0.01 sec)

mysql> use bookflow_db3;
Database changed
mysql> create table books(book_id int auto_increment primary key,title varchar(255) not null,isbn varchar(13) not null unique,published_yr int,
constraint chk_published_yr check(published_yr<2027));
Query OK, 0 rows affected (0.05 sec)

mysql> create table members(member_id INT AUTO_INCREMENT PRIMARY KEY,full_name VARCHAR(100) NOT NULL,email VARCHAR(150) NOT NULL UNIQUE);
Query OK, 0 rows affected (0.04 sec)

mysql> describe books;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| book_id | int | NO | PRI | NULL | auto_increment |
| title | varchar(255) | NO | UNI | NULL | |
| isbn | varchar(13) | NO | UNI | NULL | |
| published_yr | int | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)

mysql> describe members;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| member_id | int | NO | PRI | NULL | auto_increment |
| full_name | varchar(100) | NO | | NULL | |
| email | varchar(150) | NO | UNI | NULL | |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.01 sec)

mysql> INSERT INTO books (title, isbn, published_yr) VALUES ('The Alchemist', '9780061122415', 1988),('Clean Code', '9780132350884', 2008),('Atomic Habits', '9780735211292', 2018);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'),('Priya Sharma', 'priya.sharma@example.com'),('Ravi Verma', 'ravi.verma@example.com');
```

```
mysql> INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'),('Priya Sharma', 'priya.sharma@example.com'),('Ravi Verma', 'ravi.verma@example.com');
Query OK, 3 rows affected (0.03 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> select * from books;
+-----+-----+-----+-----+
| book_id | title       | isbn       | published_yr |
+-----+-----+-----+-----+
| 1       | The Alchemist | 9780061122415 | 1988       |
| 2       | Clean Code   | 9780132350884 | 2008       |
| 3       | Atomic Habits | 9780735211292 | 2018       |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> select * from members;
+-----+-----+-----+
| member_id | full_name | email                |
+-----+-----+-----+
| 1         | Anil Kumar | anil.kumar@example.com |
| 2         | Priya Sharma | priya.sharma@example.com |
| 3         | Ravi Verma | ravi.verma@example.com |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> INSERT INTO books (title, isbn, published_yr) VALUES ('Fake Copy', '9780061122415', 2000);
ERROR 1062 (23000): Duplicate entry '9780061122415' for key 'books.isbn'
mysql> INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999', 2010);
ERROR 1054 (42S22): Unknown column 'published_year' in 'field list'
mysql> INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '8888888888888', 2030);
ERROR 1054 (42S22): Unknown column 'published_year' in 'field list'
mysql> INSERT INTO members (full_name, email) VALUES ('Anil Clone', 'anil.kumar@example.com');
ERROR 1062 (23000): Duplicate entry 'anil.kumar@example.com' for key 'members.email'
mysql> |
```